PAM

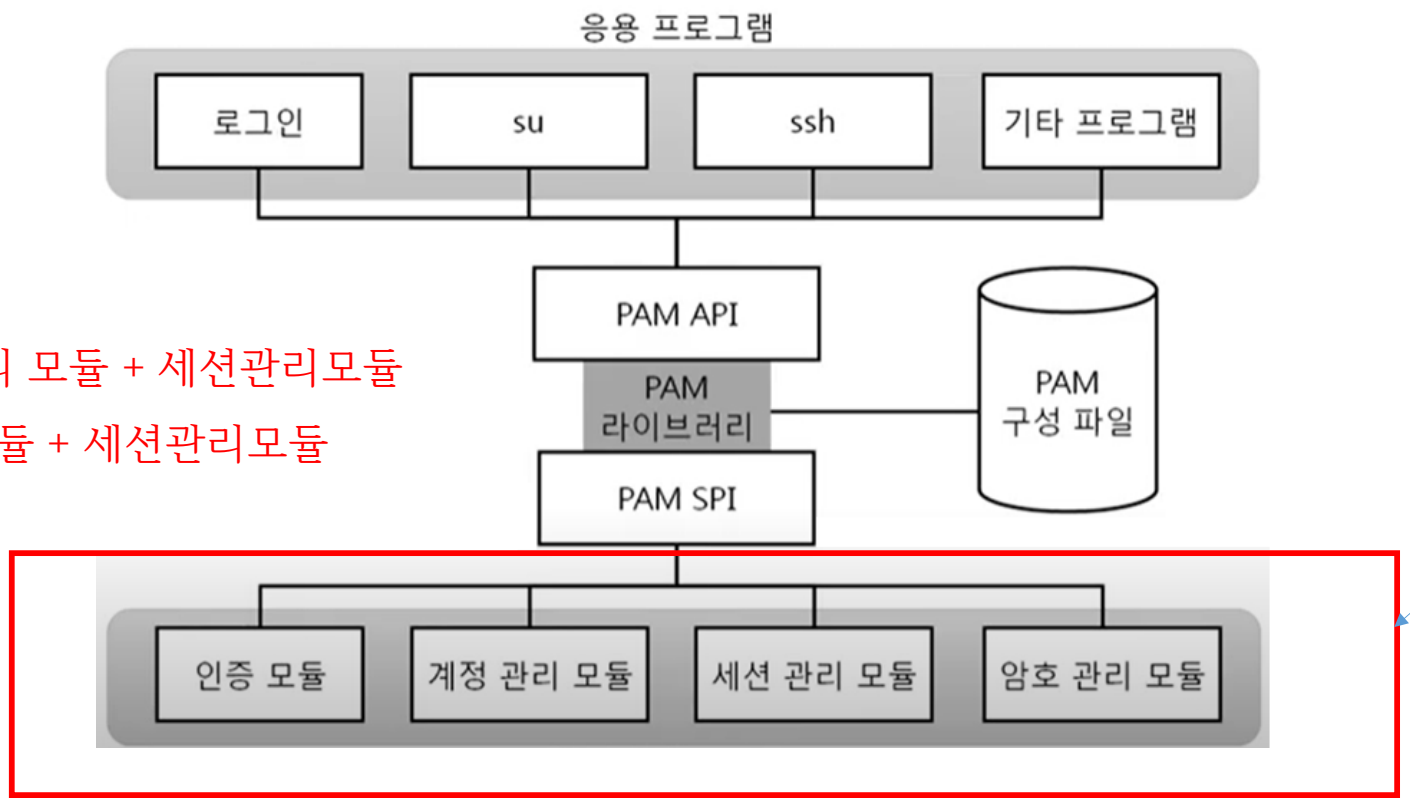
- 인증과 관련된 다양한 모듈들을 부착/제거 작업을 수행

  - 인증관련된 PAM모듈들을 인증이 필요로 하는 app에 필요할때 부착, 필요없는 경우에 제거 하는 등의 작업을 수행

  - **시스템에 공통적인 인증 방법을 제공**

ssh ➔ 계정관리 모듈 + 세션관리모듈

➔ 인증 모듈 + 세션관리모듈

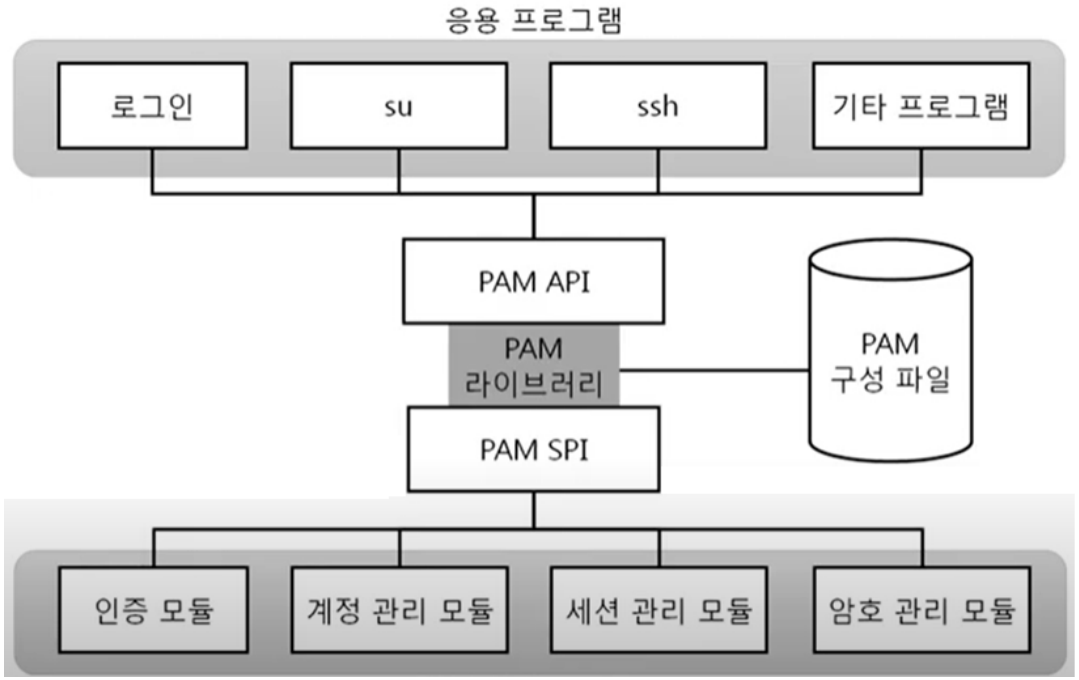


PAM 모듈

디렉터리 /etc/pam.d

- 특정 명령어와 관련된 파일들
  (예) passwd, su
- PAM을 적용한 프로그램이나
  서비스들의 설정 내용을
  저장해 둔 파일들이 모여 있는 디렉터리

```
[root@localhost pam.d]# ls
atd                  gdm-autologin          login              postlogin-ac       smtp               system-auth-ac
chfn                 gdm-fingerprint        other              ppp                smtp.postfix       systemd-user
chsh                 gdm-launch-environment passwd             remote             sshd               vlock
config-util          gdm-password           password-auth      runuser            su                 vmtoolsd
crond                gdm-pin                password-auth-ac   runuser-l          su-l               xserver
cups                 gdm-smartcard          pluto              setup              sudo
fingerprint-auth     ksu                    polkit-1           smartcard-auth     sudo-i
fingerprint-auth-ac  liveinst               postlogin          smartcard-auth-ac  system-auth
```



```
[root@localhost pam.d]#
[root@localhost pam.d]# su lg
[lg@localhost pam.d]$ su gildong
암호:
[gildong@localhost pam.d]$ su root
암호:
[root@localhost pam.d]# su gildong
[gildong@localhost pam.d]$
```

#su lg

$su root

#su gildong

$su hong

- 일반 사용자가 명령어 su를 사용시에는 패스워드를 확인하지만
- 관리자 root가 명령어 su를 사용할 경우에는 패스워드를 확인하는 절차를 갖지 않음

* 명령어 su와 관련된 인증 관련 파일

```
[root@localhost pam.d]# cat su
#%PAM-1.0
auth            sufficient      pam_rootok.so
# Uncomment the following line to implicitly trust users in the "wheel" group.
#auth           sufficient      pam_wheel.so trust use_uid
# Uncomment the following line to require a user to be in the "wheel" group.
#auth           required        pam_wheel.so use_uid
auth            substack        system-auth
auth            include         postlogin
account         sufficient      pam_succeed_if.so uid = 0 use_uid quiet
account         include         system-auth
password        include         system-auth
session         include         system-auth
session         include         postlogin
session         optional        pam_xauth.so
[root@localhost pam.d]#
```

PAM 조건이 만족되면 다른 조건들은 확인 안하고 바로 성공으로 간주

PAM 모듈 pam_rootok.so : root 사용자이면 무조건 ok를 반환

```
[root@localhost pam.d]# cat su
#%PAM-1.0
auth            sufficient      pam_rootok.so
# Uncomment the following line to implicitly trust users in the "wheel" group.
#auth           sufficient      pam_wheel.so trust use_uid
# Uncomment the following line to require a user to be in the "wheel" group.
#auth           required        pam_wheel.so use_uid
auth            substack        system-auth
auth            include         postlogin
account         sufficient      pam_succeed_if.so uid = 0 use_uid quiet
account         include         system-auth
password        include         system-auth
session         include         system-auth
session         include         postlogin
session         optional        pam_xauth.so
[root@localhost pam.d]#
```

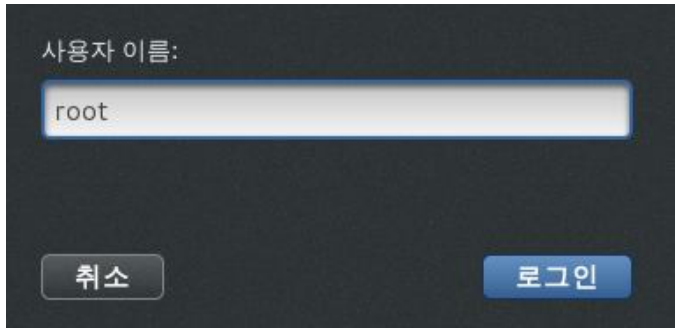
auth sufficient pam_rootok.so ➔ root 사용자가 su를 사용하면 패스워드 확인 없이 바로 다른 사용자로의 로그인 성공

/etc/pam.d 디렉터리 안에 있는 파일을 호출해서 실행 (substack와 include 유사)

```
[root@localhost pam.d]# cat su
#%PAM-1.0
auth            sufficient      pam_rootok.so
# Uncomment the following line to implicitly trust users in the "wheel" group.
#auth           sufficient      pam_wheel.so trust use_uid
# Uncomment the following line to require a user to be in the "wheel" group.
#auth           required        pam_wheel.so use_uid
auth            substack        system-auth
auth            include         postlogin
account         sufficient      pam_succeed_if.so uid = 0 use_uid quiet
account         include         system-auth
password        include         system-auth
session         include         system-auth
session         include         postlogin
session         optional        pam_xauth.so
[root@localhost pam.d]#
```

```
[root@localhost pam.d]# cat system-auth
#%PAM-1.0
# This file is auto-generated.
# User changes will be destroyed the next time authconfig is run.
auth        required      pam_env.so
auth        sufficient    pam_fprintd.so
auth        sufficient    pam_unix.so nullok try_first_pass
auth        requisite     pam_succeed_if.so uid >= 1000 quiet_success
auth        required      pam_deny.so
```

실습 1. root 로그인 불가



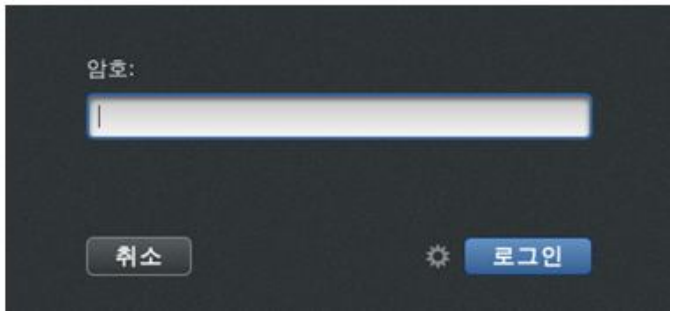
auth required pam_succeed_if.so uid >= 1000

➔ Uid가 1000 이상인 사용자들에 대해서만 성공값을 반환

➔ Pam_succeed_if.so 모듈은 조건문의 결과로 성공값을 받아 처리 함

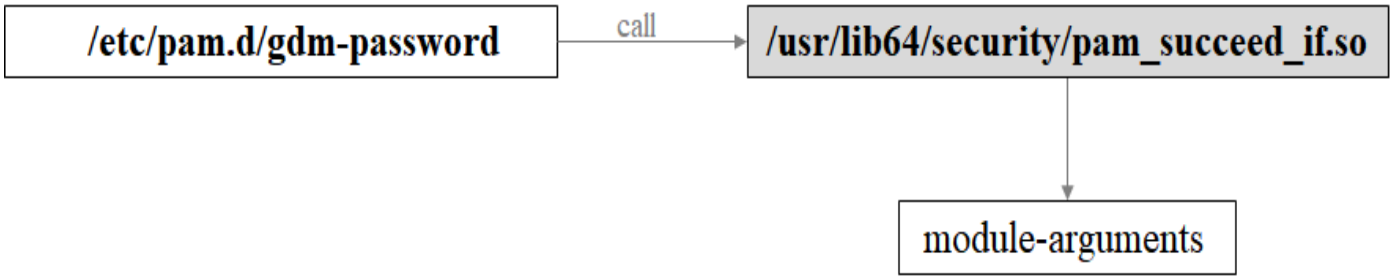
gdm-password

```
auth        required      pam_succeed_if.so   uid >= 1000
auth     [success=done ignore=ignore default=bad] pam_selinux_permit.so
auth        substack      password-auth
auth        optional      pam_gnome_keyring.so
auth        include       postlogin
```



Required

- 인증을 위해서는 반드시 이 모듈에 대한 실행결과가 성공이 되어야 함
  실패 시 나머지 모듈에 대해서도 테스트를 계속 진행하지만 인증실패가 됨



gdm-password

```
auth        required      pam_succeed_if.so   uid >= 1000
auth     [success=done ignore=ignore default=bad] pam_selinux_permit.so
auth        substack      password-auth
auth        optional      pam_gnome_keyring.so
auth        include       postlogin
```

실습 1. root 로그인 불가

- 시나리오 1. root로 바로 로그인 시
  ➔ root 패스워드만 알면됨

- 시나리오 2. root로 바로 로그인은 못하고 일반 사용자로 로그인 후 su로 root 전환 시
  ➔ 일반사용자 ID/Password
  ➔ root 패스워드 확인

wheel group

```
[root@localhost pam.d]# cat /etc/group | grep wheel
wheel:x:10:gildong,lg
[root@localhost pam.d]#
```

usermod –G 10 [사용자명]

```
[root@localhost pam.d]# cat /etc/group | grep wheel
wheel:x:10:
[root@localhost pam.d]# usermod -G 10 root
[root@localhost pam.d]# usermod -G 10 lg
[root@localhost pam.d]# usermod -G 10 gildong
[root@localhost pam.d]#
[root@localhost pam.d]# cat /etc/group | grep wheel
wheel:x:10:root,lg,gildong
[root@localhost pam.d]#
```

실습 2. su 사용자 제한

❶ wheel 그룹에 신뢰할 수 있는 사용자 추가

```
[root@localhost pam.d]# cat /etc/group | grep wheel
wheel:x:10:gildong,lg
[root@localhost pam.d]#
```

❸ 실행 결과 확인

```
[root@localhost pam.d]# su gildong
[gildong@localhost pam.d]$ su lg
[lg@localhost pam.d]$ su hong
[hong@localhost pam.d]$ su gildong
암호:
[gildong@localhost pam.d]$
```

❷ auth sufficient pam_wheel.so trust use_uid

- wheel 그룹에 속해 있는 사용자들은 신뢰할 수 있는 사용자로 간주하여 명령어 su 사용 시 패스워드 확인 없이 수행

sufficient
성공이면 아래 조건들을 테스트하지 않음

```
su
#%PAM-1.0
auth       sufficient    pam_wheel.so trust use_uid
auth       sufficient pam_rootok.so
# Uncomment the following line to implicitly trust users in the "wheel" group.
#auth      sufficient pam_wheel.so trust use_uid
# Uncomment the following line to require a user to be in the "wheel" group.
#auth      required   pam_wheel.so use_uid
auth       substack   system-auth
auth       include        postlogin
account        sufficient pam_succeed_if.so uid = 0 use_uid quiet
account        include        system-auth
password   include        system-auth
session        include        system-auth
session        include        postlogin
session        optional   pam_xauth.so
```

sufficient

- 모듈의 실행 결과가 '성공'일 경우 같은 interface에 대한 test는 더 이상 수행하지 않고 끝남

실습 3. su 사용자 제한

❶ wheel 그룹에 속해 있는 사용자 확인

```
[root@localhost pam.d]# cat /etc/group | grep wheel
wheel:x:10:root,gildong,lg
[root@localhost pam.d]#
```

*root도 포함되어 있어야 함

❸ 실행 결과 확인

```
[root@localhost pam.d]# su gildong
[gildong@localhost pam.d]$ su lg
암호:
[lg@localhost pam.d]$ su gildong
암호:
[gildong@localhost pam.d]$ su hong
암호:
[hong@localhost pam.d]$ su root
암호:
su: 권한 부여 거부
[hong@localhost pam.d]$
```

*hong은 명령어 su를 사용하지 못함

auth required pam_wheel.so use_uid

- Wheel 그룹에 속해 있는 사용자들만이 명령어 su를 사용할 수 있음
- 명령어 su를 사용하더라도 패스워드 확인 절차는 필요함

su

```
#%PAM-1.0
auth       required   pam_wheel.so use_uid
auth       sufficient pam_rootok.so
# Uncomment the following line to implicitly trust users in the "wheel" group.
#auth      sufficient pam_wheel.so trust use_uid
# Uncomment the following line to require a user to be in the "wheel" group.
#auth      required   pam_wheel.so use_uid
auth       substack   system-auth
auth       include        postlogin
account        sufficient pam_succeed_if.so uid = 0 use_uid quiet
account        include        system-auth
password   include        system-auth
session        include        system-auth
session        include        postlogin
session        optional   pam_xauth.so
```

```
[root@localhost network-scripts]# pwd
/etc/sysconfig/network-scripts
[root@localhost network-scripts]# ls -l
합계 212
-rw-r--r--. 1 root root   326  5월 11 18:58 ifcfg-eno16777736
-rw-r--r--. 1 root root   254  4월  3  2014 ifcfg-lo
```

ifcfg-eno16777736

```
HWADDR="00:0C:29:01:D5:53"
TYPE="Ethernet"
BOOTPROTO="dhcp"
DEFROUTE="yes"
PEERDNS="yes"
PEERROUTES="yes"
IPV4_FAILURE_FATAL="no"
IPV6INIT="yes"
IPV6_AUTOCONF="yes"
IPV6_DEFROUTE="yes"
IPV6_PEERDNS="yes"
IPV6_PEERROUTES="yes"
IPV6_FAILURE_FATAL="no"
NAME="eno16777736"
UUID="df47ff92-b546-4eab-be00-4ed39658e3d6"
ONBOOT="yes"
```

#systemctl restart network

#ifconfig

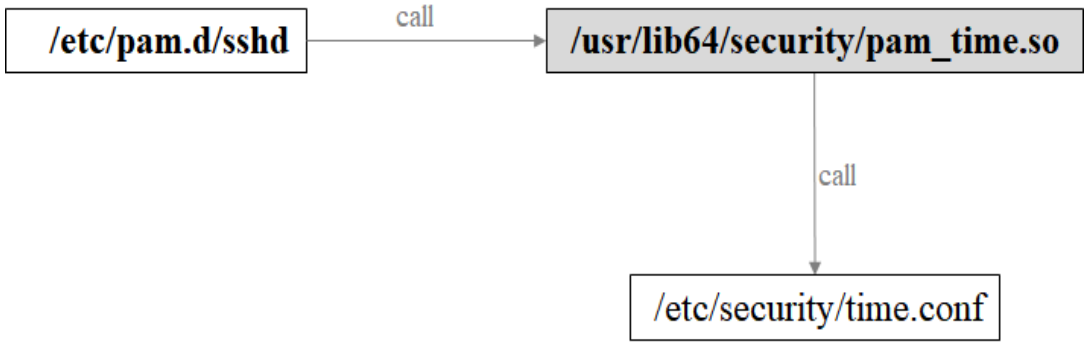
실습 4. 특정 사용자의 ssh 로그인 시간 제한

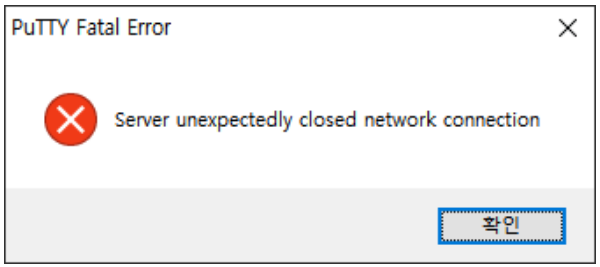
```
root@localhost:~
login as: root
root@192.168.10.129's password:
Last login: Fri May 12 03:32:53 2023
[root@localhost ~]#
```

```
[root@localhost pam.d]# cat /etc/pam.d/sshd
#%PAM-1.0
account     required      pam_time.so
auth        substack      password-auth
auth        include       postlogin
```

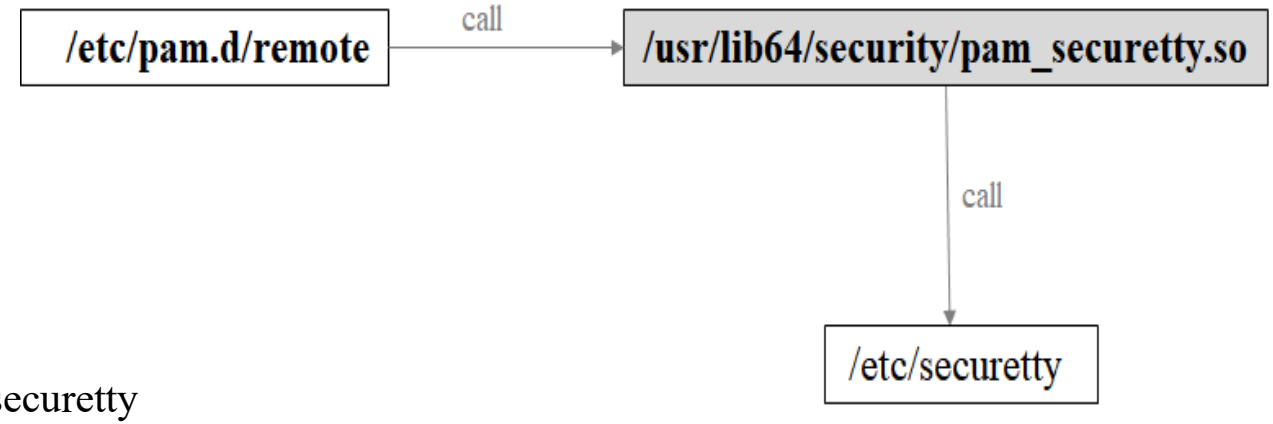
```
*time.conf
*
#
# End of example file.
#

sshd;*;root;!WK0800-17:00
```





실습 5. Telnet Remote Access 제어



#touch /etc/securetty

```
[root@localhost pam.d]# cat remote -nu
     1  #%PAM-1.0
     2  auth       required     pam_securetty.so
     3  auth       substack     password-auth
     4  auth       include      postlogin
     5  account    required     pam_nologin.so
     6  account    include      password-auth
```

실습 6. 로그인 잠금 정책 설정(콘솔로그인)

```
[root@localhost pam.d]#
[root@localhost pam.d]# pam_tally2 -u gildong
Login           Failures Latest failure     From
gildong              0
[root@localhost pam.d]# pam_tally2 -u lg
Login           Failures Latest failure     From
lg                   0
[root@localhost pam.d]# pam_tally2 -u root
Login           Failures Latest failure     From
root                 0
[root@localhost pam.d]#
```

```
#%PAM-1.0
# This file is auto-generated.
# User changes will be destroyed the next time authconfig is run.
auth        required      pam_env.so
auth        required      pam_tally2.so deny=3 unlock_time=60
auth        sufficient    pam_fprintd.so
auth        sufficient    pam_unix.so nullok try_first_pass
auth        requisite     pam_succeed_if.so uid >= 1000 quiet_success
auth        required      pam_deny.so

account     required      pam_unix.so
account     required      pam_tally2.so
account     sufficient    pam_localuser.so
account     sufficient    pam_succeed_if.so uid < 1000 quiet
account     required      pam_permit.so
```

auth required pam_tally2.so deny=3 unlock=60

account required pam_tally2.so

```
[root@localhost pam.d]#
[root@localhost pam.d]# su gildong
[gildong@localhost pam.d]$ su root
암호:
su: 인증 실패
[gildong@localhost pam.d]$ su lg
암호:
su: 인증 실패
[gildong@localhost pam.d]$ su lg
암호:
su: 인증 실패
[gildong@localhost pam.d]$ su lg
암호:
su: 인증 실패
[gildong@localhost pam.d]$ su lg
4 로그인 실패로 인해 계정이 잠김
암호:
su: 인증 실패
[gildong@localhost pam.d]$ su root
암호:
[root@localhost pam.d]# pam_tally2 -u gildong
Login           Failures Latest failure     From
gildong             0
[root@localhost pam.d]# pam_tally2 -u lg
Login           Failures Latest failure     From
lg                  4    05/12/23 05:40:13  pts/0
[root@localhost pam.d]# pam_tally2 -u root
Login           Failures Latest failure     From
root                0
```

#su gildong

$su lg

$su lg

$su lg

$su root

#pam_tally2 –u gildong

#pam_tally2 –u lg

#pam_tally2 –u root

count reset

#pam_tally2 –r –u lg

```
[root@localhost pam.d]#
[root@localhost pam.d]# pam_tally2 -u lg
Login           Failures Latest failure     From
lg                  4    05/12/23 05:40:13  pts/0
[root@localhost pam.d]#
[root@localhost pam.d]# pam_tally2 -r -u lg
Login           Failures Latest failure     From
lg                  4    05/12/23 05:40:13  pts/0
[root@localhost pam.d]#
[root@localhost pam.d]# pam_tally2 -u lg
Login           Failures Latest failure     From
lg                  0
[root@localhost pam.d]#
```

실습 7. remote access 시 (sshd, telnet, x-window)

```
[root@localhost pam.d]#
[root@localhost pam.d]# pam_tally2 -u gildong
Login           Failures Latest failure     From
gildong             0
[root@localhost pam.d]# pam_tally2 -u lg
Login           Failures Latest failure     From
lg                  0
[root@localhost pam.d]# pam_tally2 -u root
Login           Failures Latest failure     From
root                0
[root@localhost pam.d]#
```

password-auth

```
#%PAM-1.0
# This file is auto-generated.
# User changes will be destroyed the next time authconfig is run.
auth        required      pam_env.so
auth        required      pam_tally2.so deny=3 unlock_time=1200
auth        sufficient    pam_unix.so nullok try_first_pass
auth        requisite     pam_succeed_if.so uid >= 1000 quiet_success
auth        required      pam_deny.so

account     required      pam_unix.so
auth        required      pam_tally2.so
account     sufficient    pam_localuser.so
account     sufficient    pam_succeed_if.so uid < 1000 quiet
account     required      pam_permit.so
```

auth required pam_tally2.so deny=3 unlock=60

account required pam_tally2.so

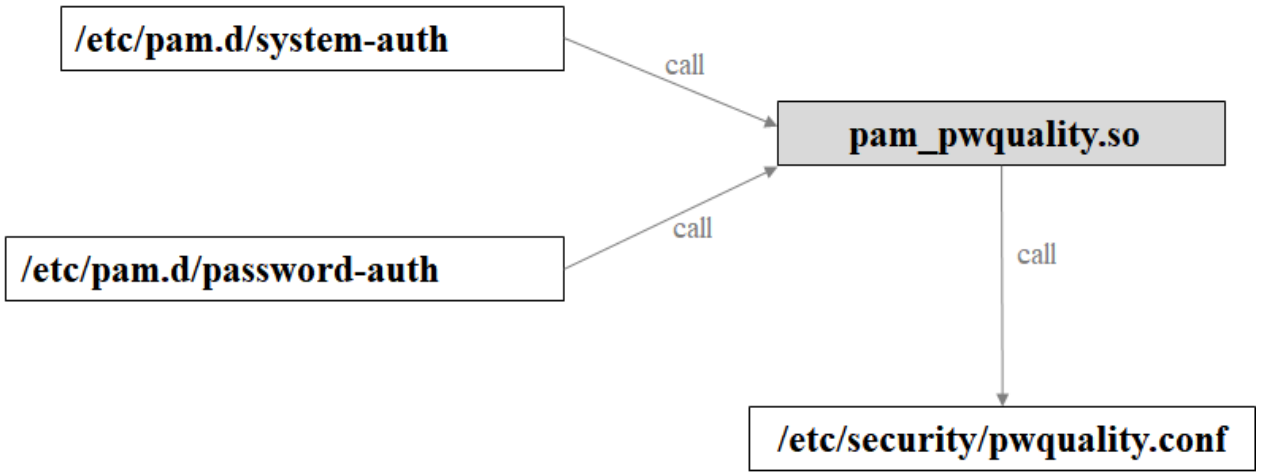
실습 7. sshd 에만 적용

*sshd

```
#%PAM-1.0
auth       required     pam_tally2.so deny=3 unlock_time=1200
auth       required    pam_sepermit.so
auth        substack     password-auth
auth        include      postlogin
account     required     pam_tally2.so
account     required     pam_nologin.so
account     include      password-auth
password    include      password-auth
# pam_selinux.so close should be the first session rule
session     required     pam_selinux.so close
session     required     pam_loginuid.so
```

실습 8. 패스워드 복잡성 검사

리눅스 6 이상버전부터는 system-auth, password-auth를 분리하여 관리
리눅스 6 이상버전부터는 pam_tally.so, pam_tally2.so 둘다 적용가능



/etc/pam.d/system-auth	로컬 로그인 (콘솔 로그인) 일반 계정에서 root 계정 전환 시(su)
/etc/pam.d/password-auth	원격 로그인 (SSHD)